

HDOC - MENU DRIVEN C PROGRAM

Increasing Patterns Using Nested Loops

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Problem Statement

Write a C program that displays a menu and uses a switch statement to print the selected increasing pattern. Use nested loops without arrays. Each pattern is implemented in a separate parameterless function with no return value.

1. Increasing right-angled triangle using asterisks.
2. Increasing row-number triangle.
3. Increasing natural-number pattern.
4. Consecutive-number triangle.
5. Increasing alphabet pattern ($1 \leq n \leq 26$).
6. Exit the program.

Algorithm

1. Start the program.
2. Display the menu of five patterns and Exit.
3. Read the user choice.
4. Use switch-case to call the selected parameterless function.
5. Inside the selected function, read n and use nested loops to print the required pattern.
6. For alphabet pattern, accept n only from 1 to 26.
7. After printing, show the menu again.
8. If choice is 6, terminate the program. For any other invalid choice, display an error message.
9. Stop.

C Program

```
#include <stdio.h>

void starTriangle();
void rowNumberTriangle();
void naturalNumberPattern();
void consecutiveTriangle();
void alphabetPattern();

int main() {
    int choice;

    do {
        printf("\n1. Increasing right-angled triangle\n");
        printf("2. Increasing row-number triangle\n");
        printf("3. Increasing natural-number pattern\n");
        printf("4. Consecutive-number triangle\n");
        printf("5. Increasing alphabet pattern\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
```

```

        case 1: starTriangle(); break;
        case 2: rowNumberTriangle(); break;
        case 3: naturalNumberPattern(); break;
        case 4: consecutiveTriangle(); break;
        case 5: alphabetPattern(); break;
        case 6: printf("Program ended.\n"); break;
        default: printf("Invalid choice! Try again.\n");
    }
} while (choice != 6);

return 0;
}

void starTriangle() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("* ");
        printf("\n");
    }
}

void rowNumberTriangle() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("%d ", i);
        printf("\n");
    }
}

void naturalNumberPattern() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}

void consecutiveTriangle() {
    int n, i, j, num = 1;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("%d ", num++);
        printf("\n");
    }
}

void alphabetPattern() {
    int n, i, j;
    printf("Enter number of rows (1-26): ");
    scanf("%d", &n);
    if (n < 1 || n > 26) {
        printf("Invalid number of rows!\n");
        return;
    }
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("%c ", 'A' + j - 1);
        printf("\n");
    }
}

```

Test-Case Table

Test	Choice	Input n	Expected result	Status
1	1	4	4-row star triangle	Pass
2	2	4	1; 2 2; 3 3 3; 4 4 4 4	Pass
3	3	4	Rows print 1 through row number	Pass
4	4	4	Consecutive numbers 1	Pass

			to 10	
5	5	4	A; A B; A B C; A B C D	Pass
6	7	-	Invalid choice message	Pass
7	6	-	Program ends	Pass

Sample Output

```

1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit

```

```

Enter your choice: 1
Enter number of rows: 4
*
* *
* * *
* * * *

```

```

Enter your choice: 2
Enter number of rows: 4
1
2 2
3 3 3
4 4 4 4

```

```

Enter your choice: 3
Enter number of rows: 4
1
1 2
1 2 3
1 2 3 4

```

```

Enter your choice: 4
Enter number of rows: 4
1
2 3
4 5 6
7 8 9 10

```

```

Enter your choice: 5
Enter number of rows (1-26): 4
A
A B
A B C
A B C D

```

```

Enter your choice: 6
Program ended.

```

Prepared for HDOC practical submission